

How ML model development Process

PART – 1: Analysis

- *Generally we have a data, the data has both input and output columns*
 - *Input columns denoted with X , output column denoted with y*
 - *We divide data into two parts*
 - *Train data*
 - *Test data*
 - *For example we have a data with 1000 observations*
 - *we divide data into 80: 20 ratio or 75: 25 ratio or 90: 10 ratio or 70: 30 ratio*
 - *80: 20 means 80% Train data 20% Test data*
 - *By default python code divide data into 75: 25*
 - *For 80: 20 out of 1000 observations*
 - *800 observations = Train data*
 - *200 observations = Test data*
 - *Train data: Data which is used to train the model or develop the model*
 - *Test data : Data which is used to test the model*
 - *Train data also has both input data and output column data*
 - *Input data = X_{train}*
 - *Output column data = Y_{train}*
 - *Test data also has both input data and output column data*
 - *Input data = X_{test}*
 - *Output column data = Y_{test}*
 - *The model is Developed by using X_{train} and y_{train}*
 - *Once Model is developed we will pass the only X_{test} (test input), this will give predictions and that predictions we called as $y_{predictions}$*
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- *This $y_{predictions}$ compare with y_{test} then we calculate accuracy of the model*

PART – 2: Analysis

- *Data is divided into two parts, train data and test data*
- *train data : X_{train} and y_{train}*
- *Test data: X_{test} and y_{test}*
- *Model developed by using X_{train} and y_{train}*

After model develop we want to test the mode in two ways

- we will pass the X_{train} only
- Model will take X_{train} only, model will give some outpt
- That output we always called as $y_{predictions}$ only
- Now this $y_{predictions}$ will compare with y_{train}
- Generally accuracy should be high
- Here the error is called as : **Train error**
- Generally Train error should be low
- If Train error is high it is called as : **UNDERFIT Model**

x y

X_train Y_train

1 1

2 4

3 9

4 16 ===== Y= x*x

x=3 y_prediction=19 vs y_train : train error

X_test y_test

5 25

6 36 x=6 y_pred=34 vs y_test=36 : test error

After model develop we want to test the mode in two ways

- we will pass the X_{test} only
- Model will take **X_{test} only**, model will give some outpt
- That output we always called as $y_{predictions}$ only
- Now this $y_{predictions}$ will compare with y_{test}
- Generally accuracy should be high
- Here the error is called as : **Test error**
- Generally **Test error should be low**
- **If Test error is high it is called as : Overfit Model**

Data : X and y

train data : X_{train} and y_{train}

Test data : X_{test} and y_{test}

Model developed by: X_{train} and y_{train}

1) you are passing only X_{train} : $y_{prediction}$ vs y_{train} : Train error

2) you are passing only X_{test} : $y_{prediction}$ vs y_{test} : Test error

we have three types of fittings

- Under fit : train error is very high (No need of test)
- Over fit: Train error is low and Test error is Very huge

The model is just mugup

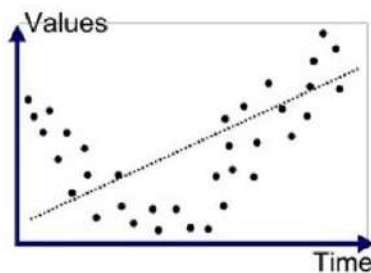
The model is Not understanding the patterns

Im giving 5qns and 5 answrs : Mugup

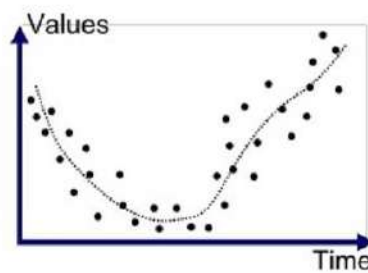
Im giving the 6thqns =====

= you are not giving the correct answr: Test error is high

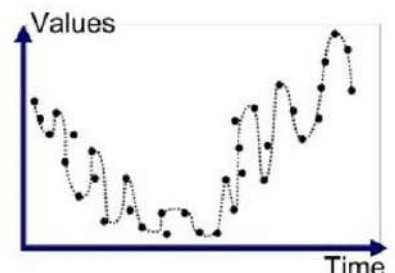
Normal fit: Train error low ===== Test error also low



Underfitted



Good Fit/Robust



Overfitted

